

Wemoji: Towards Designing Complementary Communication Systems in Augmented Reality

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ABSTRACT

Augmented Reality (AR) can enable new forms of self-expression and communication. However, little is known about how AR experiences should be designed to complement face-to-face communication. We present an initial set of insights derived from the iterative design of a mobile AR app called *Wemoji*. It enables people to emote or react to one another by spawning visual AR effects in a shared physical space. As an additional communication medium, it can help add volume and dimension to what is exchanged. We outline a design space for AR complementary communication systems, and offer a set of insights from initial testing that points towards how AR can be used to enhance same-time same-place social interactions.

CCS CONCEPTS

• Human-centered computing → Mixed / augmented reality.

KEYWORDS

Augmented Reality, Communication

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1 INTRODUCTION

Augmented Reality (AR) is steadily becoming more accessible and ubiquitous. As we approach the vision of “wearing” AR [9], an opportunity opens up for people to leverage this technological medium for self-expression and communication [10]. While technology often helps connect people in different time zones and places, AR has the potential to supplement and augment our face-to-face social interactions in a *complementary communication system* [4].

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Figure 1: Wemoji app screenshots. Left: Speaker triggers an AR light bulb effect during a speech. Middle: Speaker during the audience's AR reaction. Right: Speaker's view of the audience as they trigger AR confetti. The yellow orb avatars denote the tracked positions of the smartphones.

It can serve as an additional medium for communication, adding volume and dimension to what is exchanged.

Many prior works investigate AR in social contexts. For instance, several AR games help initiate and nurture positive social interactions [2, 3, 8, 15]. AR can make environments a conduit for messages, such as in Augment-able Reality by Rekimoto et al. [12]. More recently, advancements in face- and body-tracking enable people to be the focal point for augmentation. Toolkits such as Body LayARs by Pohl et al. [11] make it easier to create custom, platform-independent body-based AR experiences. Researchers are also actively investigating how AR effects should be designed for things like self-expression, such as with the EMOTE system by Şemsioğlu et al. [13]. However, while past works have explored AR for gaming or the expression of one's own state, we extend research in social AR by focusing on enhancing bi-directional social exchanges.

We explore this space with the iterative design and development of a multi-user mobile AR application called *Wemoji* (shown in Fig. 1). *Wemoji* adds an additional dimension to social exchanges by empowering users to trigger visual AR effects in the shared space. This can be done with the interaction methods depicted in Fig. 3. Users can *emote* by spawning an effect that appears adjacent to them (see Fig. 3-a), or *react* by triggering an effect to appear next to another person in a social exchange (see Fig. 3-b). Our contributions are (1) a working prototype of an AR application for the real-time enhancement of face-to-face social interactions, (2) an outline of design factors for AR complementary communication systems, and (3) an early set of insights to inform future research in this domain.

2 SYSTEM OVERVIEW

Wemoji is a multi-person mobile AR application, built to augment same-time, same-place interactions as per the CSCW matrix [1, 6]. We implemented the prototype based on the multi-user Connected Lens capabilities of Lens Studio. The app enables people to add an additional layer of expression to their real-world conversations by spawning AR effects. To start, a user launches the *Wemoji* Lens, scans the environment, and invites their friends to join via Snapchat (we tested 5 concurrent users, but up to 64 are supported). On entering, each person's device position is tracked. The 3D position can be denoted with avatars for debugging purposes, as can be seen in Fig. 1. Users can perform simple gestures to *emote* or *react* as described in Fig. 3 and shown in Fig. 1. The design and implementation of this Lens accounts for several design factors.

2.1 Interaction & Agency

Prior to implementing a technical application, we experimented with low-fidelity prototyping. We stuck emojis to the ends of Popsicle sticks (as shown in Fig. 2-left), and attempted to converse while also using these as communicative tools (see Fig. 2-right). We noticed that in a face-to-face conversation, the ability to present emojis was fun and encouraged active participation. However, it was also perceived to be more cognitively taxing. It was difficult to both control and monitor a new stream of information in addition to attending to standard verbal and non-verbal cues. Therefore, we limit the options in the app to a shortlist and make them simple and quick to access. In the future, it may be possible to automatically extract information (e.g., from one's tone of voice or facial expression) to either replace manual selection, or dynamically curate the list of options. However, detection accuracy remains a hurdle [7], and people often like unique control over their expressions [14].

2.2 Content Type

One major consideration is what should be expressed through a complementary AR communication medium. A wide range of options including one's emotions, level of agreement or disagreement, or external information sources (e.g., to support an argument) could all be relayed through AR. We began by establishing a shortlist of options that mirror the "reactions" that can be issued within video-calling apps such as Microsoft Teams and Zoom (see Fig. 1-left). To ensure ease-of-use, we used a categorical approach [5] and limited the options to positive expressions to keep conversations constructive. Since conversations constantly and rapidly evolve, we designed the AR affects to be temporary and disappear after a few seconds.

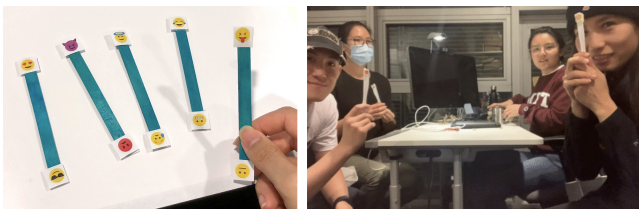


Figure 2: Low-fidelity physical prototyping. Left: Emojis stuck onto Popsicle sticks. Right: Testing the physical prototypes as communicative tools in a small group conversation.

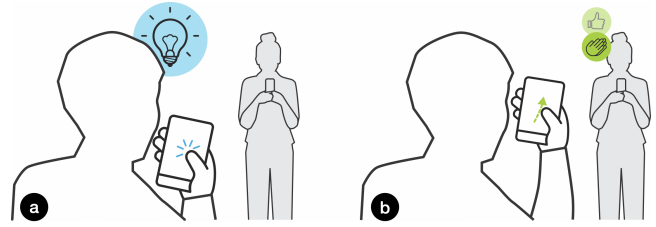


Figure 3: a) A user taps on an icon to *emote*, spawning an AR effect next to them. b) A user points the camera towards another person and *swipes-up* on an icon to *react* and send an effect to the other. Effects appear then fade over time.

2.3 Directionality & Spatiality

Wemoji uses spatial placement to distinguish between self-expression (effects spawn next to the user) and reactions towards others (effects spawn by another person). We opted for effects to simply spawn at a particular location, which keeps the sender anonymous. Alternatively, they could reveal the sender by travelling to the recipient through space. This may be desirable based on the context.

3 DESIGN REFLECTIONS

Through internal testing, we arrived at a number of insights that can be used to inform future design iterations.

- **Balance Visibility & Distraction:** Strategic positioning of AR effects within one's field of view is needed to ensure information is noticeable but not distracting. We found that spawning effects directly over one's head made them easy to miss.
- **Beyond Embodied Expressions:** The use of the thumbs up and clapping emojis were underwhelming compared to the real gestures. AR should not substitute bodily gestures but rather be used either to amplify them or support other types of expressions.
- **Amplify Immersion:** The confetti particle effect was more eye-catching than a simple party-popper image effect. 3D ambient effects with larger footprints create a more immersive experience.
- **Support Emotional Valence:** It was tempting to rapidly repeat a gesture to multiply the number of instances created and amplify one's expression. Future designs should incorporate ways to express the intensity of a sentiment.
- **Aggregation in Numbers:** The greater the number of conversation participants, the more difficult it is for someone to perceive all their states. It may be useful to aggregate expressions to maximize perception and understanding.
- **Preserve Eye Contact:** The use of a mobile device divides people's gaze between screens and others in the surrounding environment. Head-mounted displays (HMDs) could help to preserve real-world eye-contact.

4 CONCLUSION & FUTURE WORK

In this work, we presented an initial concept for an AR-based complementary communication channel. We outlined design factors and insights gained from internal testing of a functional prototype. As a next step, we would be interested in deploying this app on an AR glasses platform, and conducting user studies to investigate its potential to augment real-world co-located social exchanges.

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